

# Software Engineering

## Lab 1: Requirements Engineering & UML Use-Case Modelling

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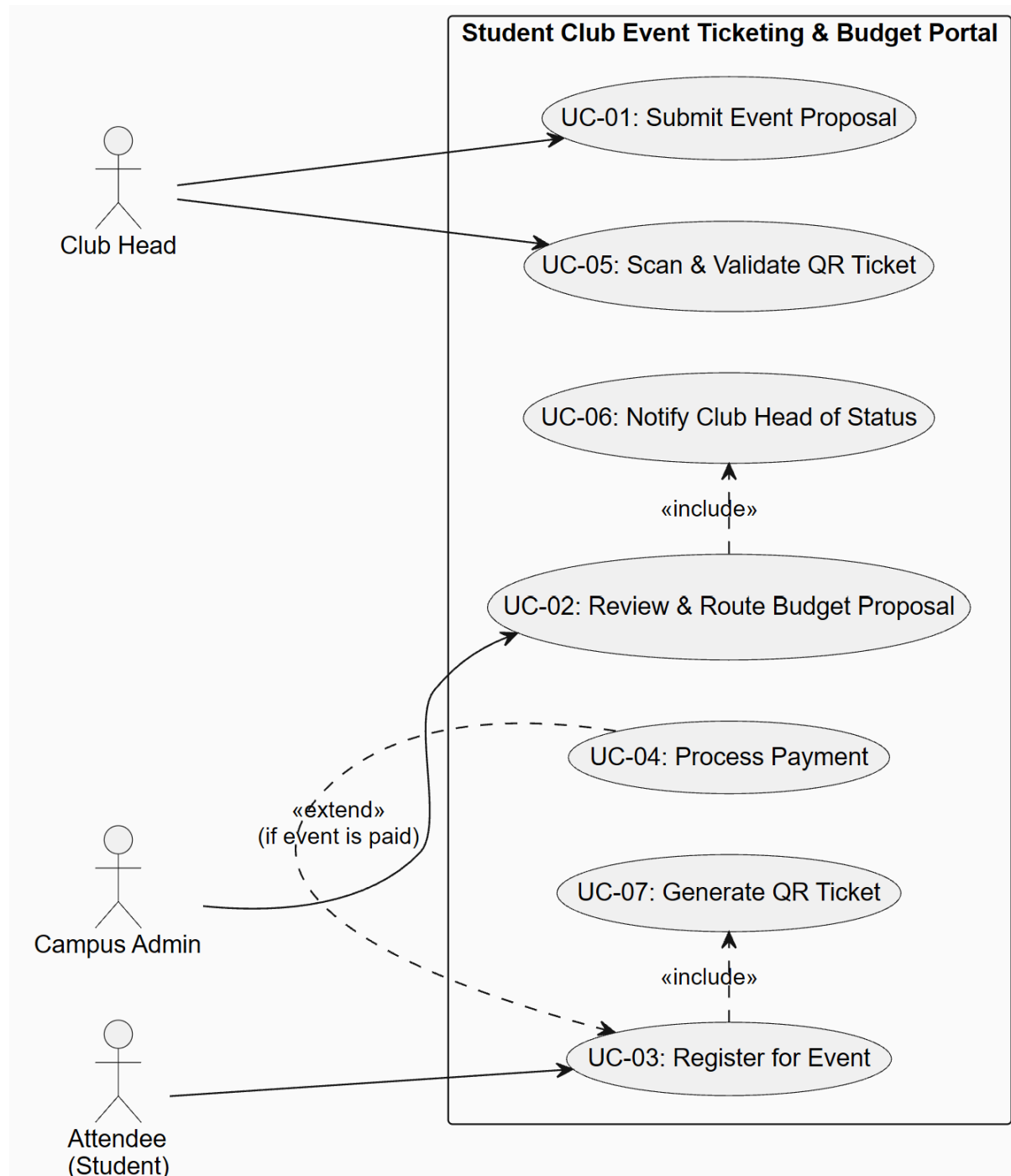
SRN: PES1UG24AM436

### Requirements Table

| <u>Req ID</u> | <u>Type</u> | <u>Description</u>                                                                                                                     | <u>Priority</u> | <u>Acceptance</u>                                                                                                                                                            | <u>Rationale</u>                                                            |
|---------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| FR-001        | Functional  | The system shall route event budget proposals through a multi-stage approval workflow (Faculty Coordinator -> Finance Office -> Dean). | High            | <b>Pass:</b> Status transitions sequentially upon digital sign-off.<br><br><b>Fail:</b> Request reaches Dean without finance coordinator sign-off.                           | Ensures strict financial oversight and compliance with university policies. |
| FR-002        | Functional  | The system shall allow a Club Head to submit an event proposal including event date, venue, expected capacity, and a line-item budget. | High            | <b>Pass:</b> Proposal successfully saves to the database and appears in the Faculty Coordinator's queue.<br><br><b>Fail:</b> Proposal submits with missing mandatory fields. | Core functionality to initiate the event planning process.                  |
| FR-003        | Functional  | The system shall generate a secure, unique QR-code ticket for each attendee upon successful event registration.                        | High            | <b>Pass:</b> System generates a scannable QR code linking to a unique attendee ID.                                                                                           | Necessary for access control and preventing ticket duplication.             |

|         |                |                                                                                                                         |        |                                                                                                                                                           |                                                                                  |
|---------|----------------|-------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|         |                |                                                                                                                         |        | <b>Fail:</b> Duplicate QR codes are generated for different users.                                                                                        |                                                                                  |
| FR-004  | Functional     | The system shall allow attendees to view approved events and register for them via a centralized dashboard.             | Medium | <p><b>Pass:</b> User clicks "Register" and is added to the attendee list.</p> <p><b>Fail:</b> Registration exceeds the maximum event capacity.</p>        | Enables the target audience to actually attend the club events.                  |
| FR-005  | Functional     | The system shall allow Campus Admins to either approve, reject, or return an event proposal for revision.               | High   | <p><b>Pass:</b> Clicking "Return" changes status to "Draft" and notifies the Club Head.</p> <p><b>Fail:</b> Status does not update upon admin action.</p> | Necessary for the administrative review cycle to function.                       |
| NFR-001 | Non-Functional | The QR-code ticket scanner API shall validate entry authenticity in under 100 ms at venue check-in points.              | High   | <p><b>Pass:</b> Benchmarking tests confirm target latency and security standards under simulated peak load</p>                                            | Prevents long queues and bottlenecks at event entrances.                         |
| NFR-002 | Non-Functional | The system shall process up to 1,000 concurrent ticket registrations without exceeding a 2-second server response time. | Medium | <p><b>Pass:</b> Load testing shows 95th percentile response times remain under 2 seconds at 1,000 concurrent users.</p>                                   | Ensures system stability during high-demand event launches (e.g., Noether Week). |

## UML Diagram



## Use-Case Flow Document

**Use Case:** UC-03 Register for Event & Receive QR Ticket **Primary Actor:** Attendee (Student) **Preconditions:**

- The Attendee is logged into the portal.
- The target event has been approved by Campus Admins and has an "Active" ticketing status.

- The event has available capacity.

**Main Success Scenario:**

1. The Attendee navigates to the "Upcoming Events" dashboard.
2. The Attendee selects a specific event and clicks the "Register" button.
3. The system checks the current registration count against the maximum venue capacity.
4. The system prompts the Attendee to confirm their university ID and contact details.
5. The Attendee confirms the details and submits the registration.
6. The system successfully registers the student and decrements the available ticket count.
7. The system generates a unique, secure QR-code ticket.
8. The system displays the QR code on the screen and emails a copy to the Attendee.
9. Use case ends successfully.